

ECA 6101 – COMPUTER NETWORKS AND CYBERSECURITY

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EXP 9 Website IP Identification and DNS Response Time Analysis using Wireshark Objectives

1. To capture and analyze DNS query and response packets using Wireshark and identify the IPv4 (A Record) and IPv6 (AAAA Record) addresses of websites.
2. To measure and evaluate the DNS response time for selected websites and verify the obtained IP addresses using the nslookup command.

OBJECTIVE 1

No.	Time	Source	Destination	Protocol	Length	Info
13803	98.344013	172.21.100.133	172.25.59.25	DNS	116	Standard query response 0x19ec A dns.google A 8.8.8.8 A 8.8.4.4
13882	99.353813	172.25.59.25	172.21.100.133	DNS	101	Standard query 0x243d A clientservices.googleapis.com
13884	99.354078	172.25.59.25	172.21.100.133	DNS	101	Standard query 0x6132 HTTPS clientservices.googleapis.com
13888	99.356432	172.21.100.133	172.25.59.25	DNS	119	Standard query response 0x243d A clientservices.googleapis.com A 142.250.206.14
13891	99.357344	172.21.100.133	172.25.59.25	DNS	160	Standard query response 0x6132 HTTPS clientservices.googleapis.com SOA ns1.google.com
17071	132.741407	172.25.59.25	172.21.100.31	DNS	86	Standard query 0x0001 PTR 31.100.21.172.in-addr.arpa
17084	132.760107	172.21.100.31	172.25.59.25	DNS	115	Standard query response 0x0001 PTR 31.100.21.172.in-addr.arpa PTR s3.saveetha.net
17085	132.769417	172.25.59.25	172.21.100.31	DNS	92	Standard query 0x0002 A www.mail.google.com saveetha.net
17087	132.820493	172.21.100.31	172.25.59.25	DNS	154	Standard query response 0x0002 No such name A www.mail.google.com saveetha.net SOA s3.saveetha.net
17088	132.821422	172.25.59.25	172.21.100.31	DNS	92	Standard query 0x0003 AAAA www.mail.google.com saveetha.net
17089	132.825982	172.21.100.31	172.25.59.25	DNS	154	Standard query response 0x0003 No such name AAAA www.mail.google.com saveetha.net SOA s3.saveetha.net
17090	132.826739	172.25.59.25	172.21.100.31	DNS	79	Standard query 0x0004 A www.mail.google.com
17093	132.916960	172.21.100.31	172.25.59.25	DNS	129	Standard query response 0x0004 No such name A www.mail.google.com SOA ns1.google.com
17095	132.917640	172.25.59.25	172.21.100.31	DNS	79	Standard query 0x0005 AAAA www.mail.google.com
17096	132.921247	172.21.100.31	172.25.59.25	DNS	129	Standard query response 0x0005 No such name AAAA www.mail.google.com SOA ns1.google.com
19465	155.225311	172.25.59.25	172.21.100.31	DNS	76	Standard query 0x987a A www.msftncsi.com
19466	155.230274	172.21.100.31	172.25.59.25	DNS	182	Standard query response 0x987a A www.msftncsi.com CNAME www.msftncsi.com.edgesuite.net CNAME a1961.g2.akamai.net A 23.205.118...

> Frame 17095: Packet, 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface \Device\NPF_{DFF30252-B7-...}

> Ethernet II, Src: AzureWaveTec_7e:a9:2d (cc:47:40:7e:a9:2d), Dst: Cisco_6d:c0:d7 (24:d5:e4:6d:c0:d7)

> Internet Protocol Version 4, Src: 172.25.59.25, Dst: 172.21.100.31

> User Datagram Protocol, Src Port: 52914, Dst Port: 53

> Domain Name System (query)

Transaction ID: 0x0005

> Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Additional RRs: 0

Queries

> www.mail.google.com: type AAAA, class IN

Name: www.mail.google.com

[Name Length: 19]

[Label Count: 4]

Type: AAAA (28) (IPv6 Address)

Class: IN (0x0001)

```
PS C:\Users\Tharun.j> nslookup www.mail.google.com
Server: s3.saveetha.net
Address: 172.21.100.31

*** s3.saveetha.net can't find www.mail.google.com: Non-existent domain
PS C:\Users\Tharun.j> |
```

OBJECTIVE 2

No.	Time	Source	Destination	Protocol	Length	Info
2647	31.105776	172.21.100.133	172.25.59.25	DNS	116	Standard query response 0xc2c2 A dns.google A 8.8.4.4 A 8.8.8.8
2648	31.105776	172.21.100.133	172.25.59.25	DNS	160	Standard query response 0x982d HTTPS dns.google SOA ns1.zdns.google
2961	36.420994	172.25.59.25	172.21.100.31	DNS	83	Standard query 0x77a3 A www.msftconnecttest.com
2976	36.459569	172.25.59.25	172.21.100.133	DNS	83	Standard query 0x77a3 A www.msftconnecttest.com
3056	37.473562	172.25.59.25	8.8.8.8	DNS	83	Standard query 0x77a3 A www.msftconnecttest.com
3057	37.483392	8.8.8.8	172.25.59.25	DNS	193	Standard query response 0x77a3 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A ...
3234	41.242913	172.21.100.31	172.25.59.25	DNS	193	Standard query response 0x77a3 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A ...
3235	41.242913	172.21.100.133	172.25.59.25	DNS	193	Standard query response 0x77a3 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A ...
3342	41.267847	172.25.59.25	172.21.100.31	DNS	74	Standard query 0x294a A assets.msn.com
3351	41.281196	172.21.100.31	172.25.59.25	DNS	262	Standard query response 0x294a A assets.msn.com CNAME assets.msn-com-world-atm-default-trafficmanager.net CNAME assets.msn.com
4319	51.179534	172.25.59.25	172.21.100.31	DNS	77	Standard query 0x0e62 A wpad.saveetha.net
4320	51.485948	172.25.59.25	172.21.100.31	DNS	77	Standard query 0x1d5d A wpad.saveetha.net
4321	52.189443	172.25.59.25	172.21.100.133	DNS	77	Standard query 0x0e62 A wpad.saveetha.net
4322	52.500855	172.25.59.25	172.21.100.133	DNS	77	Standard query 0x1d5d A wpad.saveetha.net
4555	54.193196	172.25.59.25	8.8.8.8	DNS	77	Standard query 0x0e62 A wpad.saveetha.net
4575	54.503491	172.25.59.25	8.8.8.8	DNS	77	Standard query 0x1d5d A wpad.saveetha.net
4588	54.567080	8.8.8.8	172.25.59.25	DNS	77	Standard query response 0x1d5d Server: failure A wpad.saveetha.net

> Frame 4575: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface \Device\NPF_{...}

> Ethernet II, Src: AzureWaveTec_7e:a9:2d (cc:47:40:7e:a9:2d), Dst: Cisco_6d:c0:d7 (24:d5:e4:6d:c0:d7)

> Internet Protocol Version 4, Src: 172.25.59.25, Dst: 8.8.8.8

> User Datagram Protocol, Src Port: 63749, Dst Port: 53

> Domain Name System (query)

Transaction ID: 0x1d5d

> Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

> wpad.saveetha.net: type A, class IN

Name: wpad.saveetha.net

[Name Length: 19]

[Label Count: 4]

Type: A (1) (IPv4 Address)

Class: IN (0x0001)

Results

Objective 1:

DNS query and response packets were successfully captured and analyzed using Wireshark. The IPv4 (A) records and IPv6 (AAAA) records of the selected websites were identified from the DNS response packets.

Objective 2:

The DNS response time for the selected websites was measured from the captured DNS packets and the obtained IPv4/IPv6 addresses were successfully verified using the nslookup command. The results confirmed the accuracy of the DNS resolution.

